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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	Mohd. Razia Alangir Banu
Student Name	Challagulla Shravya Chowdary
Roll Number	2520030322
Branch	CSE
Course Outcome	CO-1

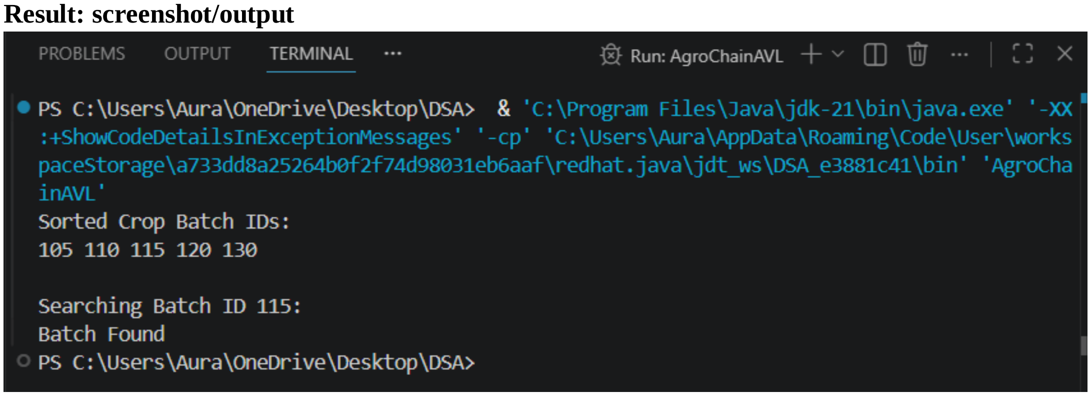
Case Study Topic: AgroChain – Smart Agriculture Supply Chain Management System

Work Done Summary:
The AgroChain Crop Inventory Management module was developed using **Binary Search Trees (BST)** and **AVL Trees** to efficiently manage agricultural crop batch records. Unique **Batch IDs** were used as keys for storing and retrieving inventory data. BST operations were implemented for inserting, searching, and deleting crop records, while AVL Tree balancing techniques (LL, RR, LR, and RL rotations) were applied to maintain optimal tree height and ensure **O(log n)** performance. Tree traversal methods were used to generate sorted crop inventory reports. The system was tested with multiple crop batch records, demonstrating efficient inventory management, faster search operations, and improved scalability for large agricultural supply chain datasets.

Implementation Details (Short)

- Used **BST and AVL Trees** to manage crop inventory records using unique **Batch IDs**.
- Implemented **insert, search, and delete** operations for crop batches.
- Applied **AVL rotations (LL, RR, LR, RL)** to keep the tree balanced.
- Used **in order traversal** to generate sorted crop inventory reports.
- Maintained efficient performance with **O(log n)** complexity for search, insertion, and deletion.
- Improved inventory management, faster record retrieval, and scalability for the AgroChain system.

Result: screenshot/output



- Crop records were stored successfully.
- AVL balancing prevented tree skewness.
- Search operation quickly located Batch ID 115.
- Sorted inventory report was generated using in order traversal.
- The system demonstrated efficient and scalable crop inventory management.

GitHub Link: **https://github.com/challagullashravya-tech/DSA-2**

NOTE: The Code and Screenshots have been uploaded in the above GitHub link.

Student Signature	Faculty Signature
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